

## SAFETY DATA SHEET

Revision Date 25-December-2021

Revision Number 4

### 1. Identification

**Product Name** Dicyclopentadiene diepoxide

**Cat No. :** AC348510000; AC348510050; AC348510250; AC348511000

**CAS-No** 81-21-0  
**Synonyms** Bicyclopentadiene diepoxide

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Manufacturer**  
Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                          |            |
|------------------------------------------|------------|
| <b>Acute oral toxicity</b>               | Category 3 |
| <b>Serious Eye Damage/Eye Irritation</b> | Category 2 |

#### Label Elements

**Signal Word**  
Danger

**Hazard Statements**  
Toxic if swallowed  
Causes serious eye irritation

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling

Do not eat, drink or smoke when using this product

Wear eye/face protection

**Response**

IF SWALLOWED: Immediately call a POISON CENTER/doctor

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Rinse mouth

If eye irritation persists: Get medical advice/attention

**Storage**

Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component                                                  | CAS-No  | Weight % |
|------------------------------------------------------------|---------|----------|
| 2,4-Methano-2H-indeno [1,2-b:5,6-b] bisoxirene, octahydro- | 81-21-0 | >95      |

### 4. First-aid measures

**General Advice**

If symptoms persist, call a physician. Show this safety data sheet to the doctor in attendance.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention. Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician. Immediately flush with plenty of water. After initial flushing, remove any contact lenses and continue flushing for at least 15 minutes. Keep eye wide open while rinsing. If symptoms persist, call a physician.

**Skin Contact**

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get medical attention. If skin irritation persists, call a physician. Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. If symptoms persist, call a physician.

**Inhalation**

Remove from exposure, lie down. Remove to fresh air. If not breathing, give artificial respiration. Get medical attention. Immediate medical attention is not required. Move to fresh air in case of accidental inhalation of vapors. If symptoms persist, call a physician.

**Ingestion**

Clean mouth with water. Get medical attention. Clean mouth with water and drink afterwards plenty of water. Do not induce vomiting without medical advice. Never give anything by mouth to an unconscious person. Consult a physician.

**Most important symptoms/effects  
Notes to Physician**

No information available.  
Treat symptomatically

## 5. Fire-fighting measures

**Suitable Extinguishing Media** Carbon dioxide (CO<sub>2</sub>). Dry chemical.

**Unsuitable Extinguishing Media** No information available

**Flash Point** No information available  
**Method -** No information available

**Autoignition Temperature** No information available

**Explosion Limits**

**Upper** No data available

**Lower** No data available

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

### Specific Hazards Arising from the Chemical

Keep product and empty container away from heat and sources of ignition.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
3

**Flammability**  
1

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

**Personal Precautions** Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**Environmental Precautions** Do not flush into surface water or sanitary sewer system. Prevent further leakage or spillage if safe to do so. Prevent product from entering drains. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up** Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal.

## 7. Handling and storage

**Handling** Avoid contact with skin and eyes. Do not breathe dust. Pay attention to flashback.

**Storage.** Keep containers tightly closed in a cool, well-ventilated place. Incompatible Materials. Strong oxidizing agents.

## 8. Exposure controls / personal protection

**Exposure Guidelines** This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**  
**Hand Protection**Goggles  
Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Nitrile rubber | recommendations   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

No protective equipment is needed under normal use conditions.

**Environmental exposure controls**

No information available.

**Hygiene Measures**

When using do not eat, drink or smoke. Provide regular cleaning of equipment, work area and clothing.

**9. Physical and chemical properties**

|                                        |                               |
|----------------------------------------|-------------------------------|
| Physical State                         | Solid                         |
| Appearance                             | Beige                         |
| Odor                                   | Odorless                      |
| Odor Threshold                         | No information available      |
| pH                                     | No information available      |
| Melting Point/Range                    | 185 - 189 °C / 365 - 372.2 °F |
| Boiling Point/Range                    | 326 °C / 618.8 °F             |
| Flash Point                            | No information available      |
| Evaporation Rate                       | Not applicable                |
| Flammability (solid,gas)               | No information available      |
| Flammability or explosive limits       |                               |
| Upper                                  | No data available             |
| Lower                                  | No data available             |
| Vapor Pressure                         | No information available      |
| Vapor Density                          | Not applicable                |
| Specific Gravity                       | No information available      |
| Solubility                             | No information available      |
| Partition coefficient; n-octanol/water | No data available             |
| Autoignition Temperature               | No information available      |
| Decomposition Temperature              | No information available      |
| Viscosity                              | Not applicable                |
| Molecular Formula                      | C10 H12 O2                    |
| Molecular Weight                       | 164.2                         |

**10. Stability and reactivity**

**Reactive Hazard** None known, based on information available

**Stability** Stable under normal conditions.

|                                         |                                                         |
|-----------------------------------------|---------------------------------------------------------|
| <b>Conditions to Avoid</b>              | Incompatible products.                                  |
| <b>Incompatible Materials</b>           | Strong oxidizing agents                                 |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ) |
| <b>Hazardous Polymerization</b>         | No information available.                               |
| <b>Hazardous Reactions</b>              | None under normal processing.                           |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

##### Component Information

| Component                                                     | LD50 Oral                | LD50 Dermal | LC50 Inhalation |
|---------------------------------------------------------------|--------------------------|-------------|-----------------|
| 2,4-Methano-2H-indeno<br>[1,2-b:5,6-b] bisoxirene, octahydro- | LD50 = 210 mg/kg ( Rat ) | Not listed  | Not listed      |

**Toxicologically Synergistic** No information available

#### Products

##### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                                                          | CAS-No  | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|--------------------------------------------------------------------|---------|------------|------------|------------|------------|------------|
| 2,4-Methano-2H-indeno<br>o [1,2-b:5,6-b]<br>bisoxirene, octahydro- | 81-21-0 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system.

**Persistence and Degradability** Persistence is unlikely

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

| Component                                                  | log Pow |
|------------------------------------------------------------|---------|
| 2,4-Methano-2H-indeno [1,2-b:5,6-b] bisoxirene, octahydro- | 1.187   |

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

UN-No UN2811  
 Proper Shipping Name consumer commodity TOXIC SOLIDS, ORGANIC, N.O.S.  
 Hazard Class 6.1  
 Packing Group III

#### TDG

UN-No UN2811  
 Proper Shipping Name Toxic solid, organic, n.o.s.  
 Hazard Class 6.1  
 Packing Group III

#### IATA

UN-No UN2811  
 Proper Shipping Name Toxic solid, organic, n.o.s.  
 Hazard Class 6.1  
 Packing Group III

#### IMDG/IMO

UN-No UN2811  
 Proper Shipping Name Toxic solid, organic, n.o.s.  
 Hazard Class 6.1  
 Packing Group III

### 15. Regulatory information

#### International Inventories

| Component                                                  | CAS-No  | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|------------------------------------------------------------|---------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| 2,4-Methano-2H-indeno [1,2-b:5,6-b] bisoxirene, octahydro- | 81-21-0 | -   | X    | X    | ACTIVE                                        | 201-334-1 | -      | -   |

| Component                                                  | CAS-No  | IECSC | KECL | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|------------------------------------------------------------|---------|-------|------|------|------|------|------|-------|-------|
| 2,4-Methano-2H-indeno [1,2-b:5,6-b] bisoxirene, octahydro- | 81-21-0 | -     | -    | X    | X    | X    | X    | X     | -     |

#### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

#### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

#### Other International Regulations

#### Authorisation/Restrictions according to EU REACH

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                                                  | CAS-No  | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|------------------------------------------------------------|---------|----------------|------------------------------|---------------------------|--------------------------------------------|
| 2,4-Methano-2H-indeno [1,2-b:5,6-b] bisoxirene, octahydro- | 81-21-0 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                                                  | CAS-No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------------------------------------------------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| 2,4-Methano-2H-indeno [1,2-b:5,6-b] bisoxirene, octahydro- | 81-21-0 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

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**Revision Date** 25-December-2021

**Print Date** 25-December-2021

**Revision Summary** This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**